

Project Proposal



Gesture Control Smart Room

PROJECT NAME	Gesture Control Smart Room
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I. PROJECT OVERVIEW

PROJECT BACKGROUND

Smart home concept is a trending and latest concept among the modern world. Since the world is now concern about the efficiency and the comfort everything converts into the smart concept rapidly. Since the initial cost is considerably high in Srilanka, this smart home concept is still not familiar among us. Our main goal is to bring this concept into Srilankan market and design gesture control system to a smart room for an affordable price.

Who we are

We are a group of engineering undergraduates who are passionate about latest technologies and their applications towards human benefit. Our group consists of both Computer engineering and Electronic engineering students who has a common objective of making things better for srilankans through smarter technology applications.

- Team leader - K.A.D.L.D.Randika
- Team members -
 - W.K.A.M.K.K.Arthanayaka
 - S.H.S.Kavinda
 - R.A.G.B.Premasiri
 - K.K.D.D.P.Jayasuriya
 - I.M.S.U.Illankoon
 - D.P.B.Athapaththu
 - H.M.J.J.Bandara

PROJECT SCOPE

In our project we are proposing to design a device to enable certain systems in a room to be controlled by gesture signals. We planned to make this device voice active. This project aims specifically the Srilankan market and therefore we have decided to focus only the systems which used in a Srilankan domestic environment. The elements we target are the light system, fan and the air conditioners up on their availability.

This device is designed to be small in size and cheap in price. And also one device is designed for all the three systems because buying a single device for all the equipment are more motivating than buying one for each.

The positive scope in our devise are as follows;

- Easy to implement
- Efficiency of controlling
- Managing all of the room devices from one place
- Affordable for anyone
- Small space is required for installing the device

2.PROJECT DESCRIPTION

As we mention before smart home concept is already in the international market. But when it comes to the local market, this smart home concept is not very much familiar with every level of people in Sri Lanka. Lack of technological knowledge, lack of uses of technological equipment Sri Lankan people are not using this kind of concept. On the other hand, people have to put large amount of money for a smart home concept. Further, in the international market the smart home concept is established only with the equipment that are specially manufactured with that feature.

When the foreign consumer needs to apply this concept for his residence he has to replace the whole equipment with the appliances which having this smart feature. Therefore, it's not very much applicable for a country like Sri Lanka.

When it comes to the local market the smart devices are existing only for the prestige crowd. And also, the existing system is not affordable. Therefore, we came up with an idea to implement a device to convert existing appliances to smart ones. The proposed device is a plug and play device which is controlled by radio signals with WIFI networks. It is designed to controlled by an application installed to the mobile phone. Only for this stage we planned to establish this concept only for the lighting system, television, air conditioner and the fan. Therefore, we bring up this concept in the name of 'GESTURE CONTROL SMART ROOM KIT' in this proposal and this device will be custom designed to meet the requirements of the local market.

3. ESTIMATING THE OUTCOME

Deliverables and benefited parties

This device will be available/delivered in open market.

In Sri Lanka the smart home concept is mostly famous with luxury level and even for them gesture control is not available still. Since one of our major intensions is to make the all levels get familiar with this smart home concept through this device. Smart Home concept is a huge step forward in the world and energy efficiency will have a positive remark with this, accessibility, safety, cost effectiveness are the achievements which we can gain. Therefore, implementing our proposed item leads the society to get familiar with this concept and through this there are lots of benefits that we can direct towards Sri Lankan economy and as well as the Sri Lanka life style.

4. HIGH LEVELL TIMELINE/SHEDULE

Time	Project Milestones
2018 - September	• Designing the project and planning the construction steps • Dividing the project into sub categories making the plan for each • Searching for the basic components which need to construct basic parts
2018 – October	• Construct the basic structure for the sub categories and develop each part.
2018 – November	• Assembling all the sub parts and construct the basic output.
2018 – December	• Looking for the further error correction • Developing the structure
201 – January	• Trying for the implementations • Testing the final construction • Doing the Final adjustments and modifications

5. BUDGET

COMPONENTS	UNIT PRICE(Rs.)	QUANTITY	SUB TOTAL(Rs.)
Relay Module	150	4	600
Fan Regulator	500	1	500
WIFI Module	600	4	2400
Arduino NANO	900	1	900
Other expenses			2000
ESTIMATE TOTAL			6400

6. NEXT STEPS

- Can implement this to apply for all electrical devices
- Can upgrade to manage other internal options other the main options which handle from a remote
- Can implement an alert system to warn to switch off when there's no one to use it – can be used as a solution for power wastage

7. ACCEPTANCE OF PROPOSAL

AUTHORIZED CLIENT SIGNATURE		AUTHORIZED CLIENT SIGNATURE	
NAME OF SIGNATORY		NAME OF SIGNATORY	
DATE OF ACCEPTANCE		DATE OF ACCEPTANCE	